

数学竞赛研讨课第一节

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1 一些常用不等式

定理 1.1: 均值不等式

对于任意的正数 $a_1, a_2, \dots, a_n$, 都有

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

证明. (1) 向前数学归纳法:

归纳假设:

设 $P(n)$ 为命题 “对于任意的正数 $a_1, a_2, \dots, a_n$, 都有

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

基例:

$n = 1$ 时, 显然成立。

假设:

假设 $P(2^k)$ 成立, 即对于任意的正数 $a_1, a_2, \dots, a_{2^k}$, 都有

$$\frac{a_1 + a_2 + \dots + a_{2^k}}{2^k} \geq \sqrt[2^k]{a_1 a_2 \dots a_{2^k}}.$$

那么对于 $n = 2^{k+1}$

$$\text{记 } A_1 = \frac{a_1 + a_2 + \dots + a_{2^k}}{2^k}, A_2 = \frac{a_{2^k+1} + \dots + a_{2^{k+1}}}{2^k}$$

$$G_1 = \sqrt[2^k]{a_1 a_2 \dots a_{2^k}}, G_2 = \sqrt[2^k]{a_{2^k+1} a_{2^k+2} \dots a_{2^{k+1}}}$$

由归纳假设知, $A_1 \geq G_1, A_2 \geq G_2$

$$\frac{a_1 + a_2 + \dots + a_{2^{k+1}}}{2^{k+1}} = \frac{A_1 + A_2}{2} \geq \sqrt{G_1 G_2} = \sqrt[2^{k+1}]{a_1 a_2 \dots a_{2^{k+1}}}$$

下证对于任何 n 都成立.

对于任意正整数 n , 取最小的 m 使得 2^m 大于等于 n , 构造 2^m 个非负实数: $a_1 = a_2 = \dots = a_n = 1, a_{n+1} = \dots = a_{2^m} = 0$ 因此, 命题 $P(n)$ 对任意正整数 n 成立. $\square$

But this immediately implies the following corollary:

推论 1.1. *Riemann's Every non-trivial zero of the Riemann ζ function has real part one-half.*

Which we can demonstrate with an example:

例题 1.1: Poincare

Consider a simply connected, closed 3-manifold. Notice that it is homeomorphic to the 3-sphere!

引理 1.1. *This is a lemma.*

命题 1.1. *This is a proposition.*

注记. This is a note.

定义 1.1: Limit of a function

If, for every $\epsilon > 0$ there exists some $\delta > 0$ such $0 < x - a < \delta$ implies $|f(x) - L| < \epsilon$ then we say that the function f has a limit of L at a and we write

$$\lim_{x \rightarrow a} f(x) = L.$$